CROWNTOUCH LOGISTICS

ENTERPRISE NETWORK DESIGN, CONFIGURATION AND IMPLEMENTATION DOCUMENTATION

Cisco Packet Tracer Project

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1. Executive Summary

CrownTouch Logistics required a reliable and secure enterprise network connecting its Lagos headquarters to branch offices in Abuja, Port Harcourt, and Ibadan. The project implements a hub-and-spoke topology in Cisco Packet Tracer using routers, multilayer switches, access switches, servers, and end-user devices.

The solution includes departmental VLANs, inter-VLAN routing, OSPF, DHCP, DNS, file-server connectivity, LACP EtherChannel, Rapid PVST+, ACL-based restrictions, SSH administration, switch port security, and NAT overload for simulated internet access. Testing confirmed successful connectivity for the tested paths, DNS resolution, NAT translations, the HR-to-Sales restriction, and recovery of OSPF after restoration of the Port Harcourt WAN link.

The project is substantially implemented and tested. Outstanding verification items are documented rather than being represented as complete.

2. Project Background and Objectives

2.1 Problem Statement

- Unreliable communication between headquarters and branch offices.
- Dependence on mobile hotspots for some cross-site communication.
- Consumer-grade ISP routers unsuitable as the core of a managed enterprise network.
- Risk of Layer 2 loops and broadcast storms caused by incorrect cabling.
- Insufficient separation of sensitive HR traffic from other departments.
- Manual IP addressing that increases configuration errors and duplicate-address risk.

2.2 Objectives

1. Connect Lagos HQ with Abuja, Port Harcourt, and Ibadan.
2. Implement a hub-and-spoke WAN architecture.
3. Separate departmental traffic using VLANs.
4. Enable inter-VLAN routing and dynamic OSPF routing.
5. Provide DHCP, DNS, and centralized file-server connectivity.
6. Enforce departmental access restrictions using ACLs.
7. Improve Layer 2 resilience using LACP EtherChannel and Rapid PVST+.
8. Secure device administration through SSH and restrict unauthorized access using port security.
9. Provide simulated internet access through HQ using NAT overload.

3. Network Architecture

The network uses Lagos as the central hub. Abuja, Port Harcourt, and Ibadan are spokes connected to HQ-R1 through serial WAN links. The headquarters core consists of two multilayer switches, MCORE1 and MCORE2, connected using two physical links bundled into an LACP EtherChannel. Access switches connect departmental PCs, printers, wireless access equipment, and servers.

3.1 Devices and Roles

Device	Role
HQ-R1	Headquarters router, WAN gateway, default route, and NAT
ABJ-R1	Abuja branch router

PH-R1	Port Harcourt branch router
IB-R1	Ibadan branch router
MCORE1	Headquarters multilayer core switch and primary VLAN gateways
MCORE2	Redundant headquarters multilayer core switch
HQ-SW1	Headquarters server access switch
HQ-SW2	Headquarters user access switch
Branch access switches	Connect branch PCs, printers, and other endpoint devices
DHCP server	Automatic IP address allocation
DNS server	Internal hostname resolution
File server	Centralized file-service connectivity
ISP-R1	Simulated upstream ISP router
INTERNET-SERVER	Simulated external destination

3.2 Physical Connectivity

- HQ-R1 connects to each branch router over a serial WAN link.
- HQ-R1 connects to the headquarters core over Ethernet.
- MCORE1 and MCORE2 are linked by two physical LACP member links.
- Access switches uplink to the headquarters core.
- Servers connect through HQ-SW1; user endpoints connect through access switches.
- HQ-R1 G0/2 connects to ISP-R1 G0/0; ISP-R1 G0/1 connects to INTERNET-SERVER.

4. IP Addressing Plan

4.1 Headquarters VLANs

VLAN	Department	Network	Primary Gateway
10	Sales	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1
30	IT / Servers	192.168.30.0/24	192.168.30.1
99	Management	192.168.99.0/24	192.168.99.1

4.2 Core Switch SVI Addresses

SVI	MCORE1	MCORE2
VLAN 10	192.168.10.1	192.168.10.2
VLAN 20	192.168.20.1	192.168.20.2
VLAN 30	192.168.30.1	192.168.30.3
VLAN 99	192.168.99.1	192.168.99.2

4.3 Server Addresses

Server	Address	Gateway
DHCP server	192.168.30.10	192.168.30.1
DNS server	192.168.30.11	192.168.30.1
File server	192.168.30.12	192.168.30.1

4.4 Branch Networks

Branch	Sales	HR	IT	Management
Ibadan	192.168.110.0/24	192.168.120.0/24	192.168.130.0/24	192.168.140.0/24
Abuja	192.168.150.0/24	192.168.160.0/24	192.168.170.0/24	192.168.180.0/24
Port Harcourt	192.168.190.0/24	192.168.200.0/24	192.168.210.0/24	192.168.220.0/24

The expected gateway is the .1 address within each branch subnet. Confirm the final VLAN IDs and gateways against the saved Packet Tracer file before submission.

4.5 WAN Addressing

Link	HQ-R1 interface/address	Branch interface/address
Lagos–Ibadan	S0/0/0 — 10.0.1.1/30	IB-R1 S0/0/0 — 10.0.1.2/30
Lagos–Abuja	S0/0/1 — 10.0.2.1/30	ABJ-R1 S0/0/0 — 10.0.2.2/30
Lagos–Port Harcourt	S0/1/0 — 10.0.3.1/30	PH-R1 S0/0/0 — 10.0.3.2/30

4.6 Simulated Internet Addressing

Device/interface	Address
HQ-R1 G0/2	203.0.113.1/30
ISP-R1 G0/0	203.0.113.2/30
ISP-R1 G0/1	198.51.100.1/24
INTERNET-SERVER	198.51.100.10/24; gateway 198.51.100.1

The 203.0.113.0/24 and 198.51.100.0/24 ranges are reserved for documentation and examples. They represent a simulated internet path, not a real public internet service.

5. Initial Device Setup and Cabling

Routers and switches were placed in the Packet Tracer workspace and named according to their roles. Serial interfaces were used for WAN links, while Ethernet links connected the HQ router, core switches, access switches, servers, ISP router, and external server.

5.1 Example Interface Configuration

```
enable
configure terminal
interface gigabitEthernet0/0
ip address 10.0.0.1 255.255.255.252
no shutdown
end
```

This is an example of the HQ-R1 LAN-facing interface. Interface numbers and IP addresses must match the actual device and topology.

```
show ip interface brief
show controllers serial
```

These commands were used to verify interface state and inspect serial controller details where needed.

6. VLAN Creation and Access-Port Configuration

6.1 VLAN Purpose and Creation

VLANs logically separate departments, reduce unnecessary broadcast traffic, and provide a basis for applying access-control policies.

```
enable
configure terminal
vlan 10
name SALES
vlan 20
name HR
vlan 30
name IT_SERVERS
vlan 99
name MANAGEMENT
end
```

6.2 HQ-SW2 Port Plan

Ports	VLAN	Department / Use
Fa0/1–Fa0/5	10	Sales
Fa0/6–Fa0/10	20	HR
Fa0/11–Fa0/15	30	IT
Fa0/16–Fa0/20	99	Management / management equipment

```
configure terminal
interface range fastEthernet0/1-5
switchport mode access
switchport access vlan 10
end
```

Equivalent commands were used for other port ranges with the appropriate VLAN IDs.

6.3 Branch Port Plan

Port	Intended use
Fa0/5	Sales
Fa0/10	HR
Fa0/15	IT
Fa0/20	Management access point
Fa0/24	Printer

7. Trunking and Inter-VLAN Routing

Trunk links carry multiple VLANs between switches where required. The core multilayer switches use Switched Virtual Interfaces (SVIs) to provide Layer 3 gateways for departmental VLANs.

```
configure terminal
ip routing
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
```

Equivalent SVI configurations were used for VLANs 20, 30, and 99. MCORE1 uses the primary gateway addresses; MCORE2 uses separate SVI addresses.

```
show interfaces trunk
show vlan brief
show ip interface brief
```

8. LACP EtherChannel and Rapid PVST+

8.1 LACP EtherChannel

Two physical links between MCORE1 and MCORE2 were bundled into Port-channel1 using LACP. The exact member interfaces should be confirmed against the final topology.

```
interface range gigabitEthernet1/0/23-24
channel-group 1 mode active
show etherchannel summary
```

The EtherChannel summary was checked and reported as operational, with LACP and bundled member ports.

8.2 Rapid PVST+

Rapid PVST+ was checked to confirm spanning-tree operation and help protect the switched network from Layer 2 loops.

```
show spanning-tree
show spanning-tree summary
```

The spanning-tree mode was reported as Rapid PVST+.

9. DHCP, DNS, and File Services

9.1 DHCP

The central DHCP server is located at 192.168.30.10. DHCP scopes supply client addresses, subnet masks, default gateways, and DNS server information. DHCP relay may be required on routed gateway interfaces where clients and the server are in different subnets; verify the final configuration.

```
ipconfig
```

The HQ Sales PC was initially found using Static addressing and was switched to DHCP. Branch Sales PCs at Ibadan, Abuja, and Port Harcourt were checked for DHCP selection and expected subnet settings.

Client	IPv4 address	Subnet mask	Gateway
HQ Sales	192.168.10.50	255.255.255.0	192.168.10.1
Ibadan Sales	192.168.110.50	255.255.255.0	192.168.110.1
Abuja Sales	192.168.150.50	255.255.255.0	192.168.150.1
Port Harcourt Sales	192.168.190.x	255.255.255.0	192.168.190.1

9.2 DNS

The DNS server address is 192.168.30.11. The hostname dns.crowntouch.local was tested from an HQ Sales PC and resolved to 192.168.30.11. The first request timed out, but a subsequent request returned the expected address.

9.3 File Server

The file server address is 192.168.30.12, with gateway 192.168.30.1. A ping from an HQ Sales PC succeeded. This confirms IP reachability but does not independently validate file-sharing permissions.

```
ping 192.168.30.12
```

10. OSPF Routing and Default Route

OSPF process 1 distributes internal routes between the headquarters and branches. The WAN links use the 10.0.1.0/30, 10.0.2.0/30, and 10.0.3.0/30 networks.

```
router ospf 1
network 10.0.0.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.99.0 0.0.0.255 area 0
```

This is the verified OSPF network statement set from MCORE1; other devices require their own appropriate statements.

10.1 Default Route and Advertisement

```
ip route 0.0.0.0 0.0.0.0 203.0.113.2
router ospf 1
default-information originate
```

HQ-R1 uses the ISP router as its default next hop and advertises the default route into OSPF. MCORE1 initially lacked a default route. After default-information originate was configured, MCORE1 learned an external OSPF default route via 10.0.0.1.

10.2 OSPF Verification

```
show ip ospf neighbor
show ip route ospf
show ip route 192.168.190.0
```

OSPF neighbors and branch routes were checked. Neighbors reached FULL, and branch routes were present.

10.3 WAN Restoration Test

The Port Harcourt serial link was temporarily shut down and then restored. After no shutdown, the interface returned to up/up, the OSPF adjacency recovered, and the Port Harcourt route reappeared. This verified recovery after restoration, not an alternate route during an outage.

11. ACL-Based Departmental Security

The intended security policy prevents HR (192.168.20.0/24) from initiating IP traffic to Sales (192.168.10.0/24), while permitting other traffic according to the remaining ACL rules.

```
access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 101 permit ip any any
```

The HR-to-Sales ping initially succeeded. After the ACL configuration was revisited, the ping failed, the HR PC could still ping its own gateway, and the ACL deny counter increased. Packet Tracer did not consistently display the inbound ACL under show ip interface vlan 20. Confirm the final attachment in the saved configuration before submission.

```
show access-lists 101
show ip interface vlan 20
ping 192.168.10.50
ping 192.168.20.1
```

12. SSH Administration and Telnet Restriction

SSH was configured for secure remote device administration. An SSH login to MCORE1 at 192.168.99.1 succeeded after the admin password was reset.

```
ssh -l admin 192.168.99.1
```

The checked VTY configuration contained transport input ssh, which restricts those VTY lines to SSH. Verify all VTY lines on every device that requires remote management.

```
line vty 0 4
transport input ssh
```

The password was reset during troubleshooting. Do not include the actual device password in a publicly distributed report.

13. Switch Port Security

HQ-SW2 access ports were configured for a maximum of two MAC addresses per port, using sticky MAC learning where supported. Port-security status was checked and reported as good.

```
interface fastEthernet0/5
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
show port-security
```

The exact configuration must match the actual final port assignments. Port security on branch access switches should also be audited if required by the project brief.

14. Simulated Internet and NAT Overload

14.1 ISP Simulation

ISP-R1 and INTERNET-SERVER were added to simulate an external network. ISP-R1 G0/0 uses 203.0.113.2/30 and G0/1 uses 198.51.100.1/24. INTERNET-SERVER uses 198.51.100.10/24 with gateway 198.51.100.1.

```
ip route 192.168.0.0 255.255.0.0 203.0.113.1
```

This static route was configured on ISP-R1 to route replies toward the internal company networks via HQ-R1.

14.2 HQ-R1 NAT Configuration

```
interface gigabitEthernet0/2
ip address 203.0.113.1 255.255.255.252
ip nat outside
no shutdown

interface gigabitEthernet0/0
ip nat inside
```

```
interface serial0/0/0
ip nat inside

interface serial0/0/1
ip nat inside

interface serial0/1/0
ip nat inside

access-list 10 permit 192.168.0.0 0.0.255.255
ip nat inside source list 10 interface gigabitEthernet0/2 overload
```

The inside interfaces represent internal LAN and WAN traffic; G0/2 is the outside interface. NAT overload allows internal addresses to share the translated outside interface address.

14.3 NAT Troubleshooting and Verification

Initially, the NAT translation table was empty and the hit counter was zero. HQ-R1 could reach the external server, but MCORE1 lacked a default route. After HQ-R1 advertised its default route into OSPF, the HQ Sales PC successfully pinged 198.51.100.10 and NAT translations appeared.

```
show ip nat statistics
show ip nat translations
```

Field	Example observed value
Inside local	192.168.10.50
Inside global	203.0.113.1
Outside local	198.51.100.10
Outside global	198.51.100.10

The HQ Sales PC and Port Harcourt branch both successfully pinged the external server. The address ranges used are documentation ranges and simulate internet access only within Packet Tracer.

15. Testing and Verification Results

No.	Test	Observed result	Status
1	HQ Sales PC to gateway	Ping succeeded	Passed
2	HQ-R1 to ISP and external server	Pings succeeded	Passed
3	HQ Sales PC to external server	Ping succeeded; NAT entries appeared	Passed
4	Port Harcourt PC to external server	Ping succeeded	Passed
5	HR PC to Sales PC	Ping failed; ACL deny counter increased	Functionally passed
6	HR PC to its gateway	Ping succeeded	Passed
7	Branch Sales DHCP	Ibadan, Abuja, and Port Harcourt settings checked	Passed
8	DNS hostname resolution	dns.crowntouch.local resolved to 192.168.30.11	Passed
9	HQ Sales to file server	Ping succeeded	Passed
10	SSH to MCORE1	Login succeeded	Passed
11	VTY SSH-only setting	transport input ssh found on checked lines	Checked
12	LACP EtherChannel	Reported operational	Passed
13	Rapid PVST+	Reported correct mode	Passed

14	HQ-SW2 port security	Reported good	Passed
15	OSPF neighbors and routes	Neighbors and branch routes verified	Passed
16	Port Harcourt WAN restoration	Interface, neighbor, and route recovered	Passed

16. Troubleshooting Summary

16.1 Static Instead of DHCP

The HQ Sales PC was initially set to Static addressing. DHCP was selected and the expected address and gateway were then used.

16.2 Missing NAT Translations

NAT rules, ACL 10, and inside/outside interfaces were checked. The missing default route on MCORE1 was identified and corrected by advertising the default route from HQ-R1 through OSPF. Subsequent traffic created NAT translations.

16.3 HR-to-Sales ACL

The HR-to-Sales ping initially succeeded. The ACL was revisited and the test later failed while the ACL deny counter increased. Final ACL attachment still requires confirmation in the saved configuration.

16.4 SSH Authentication

SSH authentication initially failed. The admin secret was reset from the device CLI and SSH login then succeeded.

17. Outstanding Requirements and Limitations

- WAN failover: the restoration test proved OSPF recovery after the link returned, not alternate connectivity during the outage. Each branch currently has a single WAN link to HQ.
- ACL attachment: observed behavior and deny counters indicate the HR-to-Sales restriction works, but the final SVI ACL attachment should be confirmed.
- SSH: MCORE1 was tested. Verify SSH-only management on all other required routers and switches.
- DHCP: Sales clients at all three branches were checked; audit all other departmental scopes and clients.
- File permissions: ping confirms IP reachability, not application-level access restrictions for HR salary files.
- Port security: HQ-SW2 was checked; audit required access ports on branch switches.
- Internet: ISP and external server are simulations, not a real ISP connection.

18. Recommendations

10. Maintain a current IP addressing and VLAN register.
11. Save backup copies of the Packet Tracer file after major milestones.
12. Verify all switch trunks carry the required VLANs.
13. Audit SSH-only access on all required devices.
14. Confirm the ACL attachment and retain verification output.
15. Test DHCP scopes for all departments and branches.

16. Add a second WAN path if genuine branch failover is required.
17. Configure and test file-service permissions for sensitive HR documents.
18. Retain command output or screenshots as evidence for each assessment requirement.

19. Conclusion

The CrownTouch Logistics Packet Tracer project implements a multi-site enterprise network connecting Lagos headquarters with Abuja, Port Harcourt, and Ibadan. The work includes departmental VLANs, multilayer switching, LACP EtherChannel, Rapid PVST+, OSPF routing, DHCP, DNS, file-server connectivity, ACL-based departmental restrictions, SSH management, port security, and simulated internet access through NAT overload.

Testing demonstrated connectivity between the tested endpoints and branches, successful DNS resolution, NAT translation, the intended HR-to-Sales restriction, SSH access to MCORE1, and OSPF recovery after the Port Harcourt WAN link was restored. Remaining tasks involve final configuration auditing and addressing the limitations listed in Section 17. The project should be described as substantially implemented and tested until those outstanding items are resolved.

Appendix A. Useful Verification Commands

```
show ip interface brief
show vlan brief
show interfaces trunk
show etherchannel summary
show spanning-tree summary
show port-security
show ip ospf neighbor
show ip route ospf
show access-lists
show ip nat statistics
show ip nat translations
show running-config
```

Appendix B. Submission Checklist

- Save the final Packet Tracer (.pkt) file.
- Update the Table of Contents in Microsoft Word.
- Confirm final device names, interface numbers, VLANs, and IP addresses against the saved topology.
- Add screenshots or command outputs if required by the lecturer.
- Do not include passwords in the report.
- Resolve or explicitly disclose the outstanding requirements in Section 17.